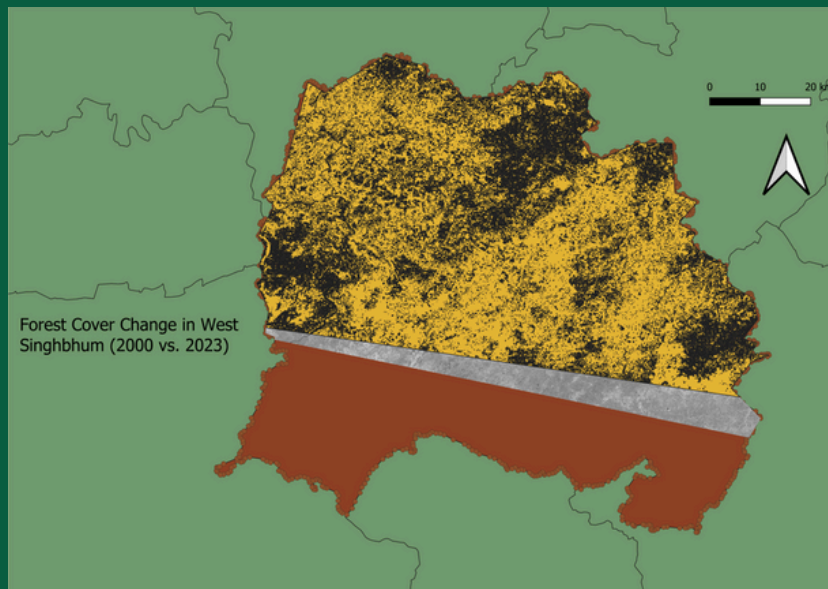


PROJECT
REPORT ON

**Title: P5 -Detecting Temporal Changes in
Forest Cover using NDVI Differencing
Technique**

A CASE STUDY OF WEST SINGHBHUM DISTRICT, JHARKHAND (2000–2023)

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PREPARED FOR :
India Space Academy
Internship

**DATE OF
SUBMISSION**
16th August 2025

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TITLE – Detecting Temporal Changes in Forest Cover using NDVI Differencing Technique

Forest cover monitoring is crucial for understanding deforestation and reforestation trends. NDVI differencing is a widely used remote sensing method that highlights temporal vegetation changes. This project applies NDVI differencing to assess forest cover change in West Singhbhum District, Jharkhand. of the year 2000 and 2023.

Objective

The main objective of this project is to assess temporal changes in vegetation cover in West Singhbhum District, Jharkhand, by applying NDVI differencing on multi-date Landsat satellite imagery (2000 vs 2023).



Study Area

West Singhbhum, a densely forested district in Jharkhand, is rich in minerals but faces a delicate balance-while mining and human activity have led to deforestation, conservation efforts have also spurred regrowth in some areas."



Data Used

Satellite Data

- Landsat-5 TM (Year 2000) – Bands 4 (NIR), 3 (Red)
- Landsat-8 OLI (Year 2023) – Bands 5 (NIR), 4 (Red)

Shapefile

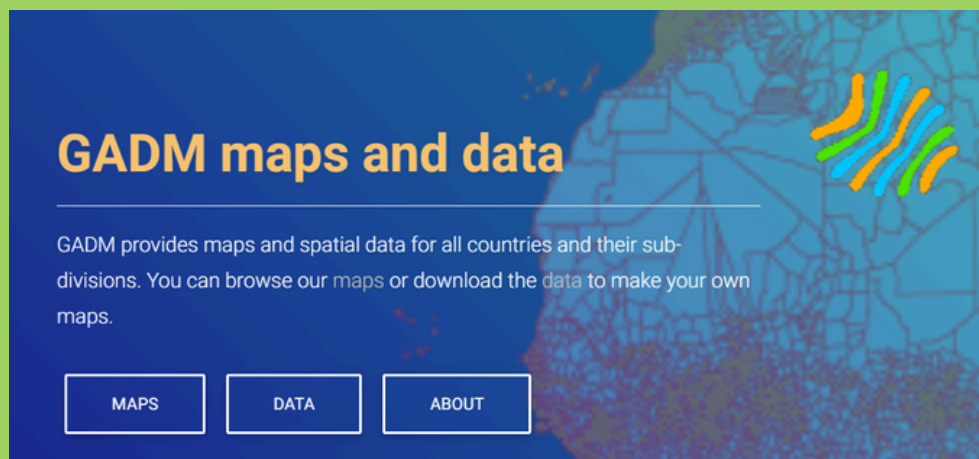
- AOI Boundary was downloaded from GADM

Sources

- USGS Earth Explorer
(<https://earthexplorer.usgs.gov/>)

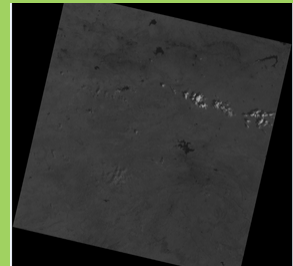
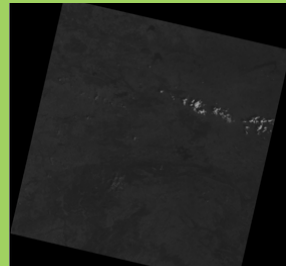
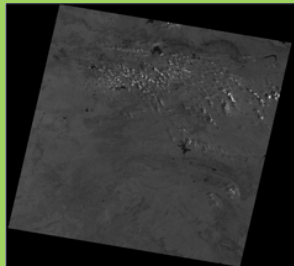
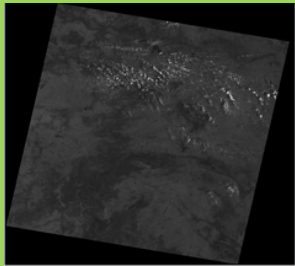


- GADM (<https://gadm.org/>)



Methodology

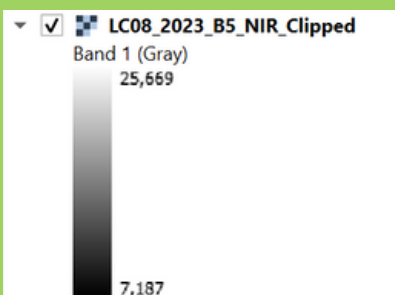
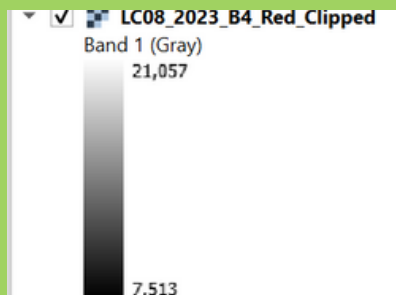
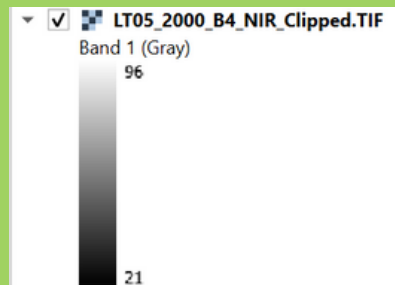
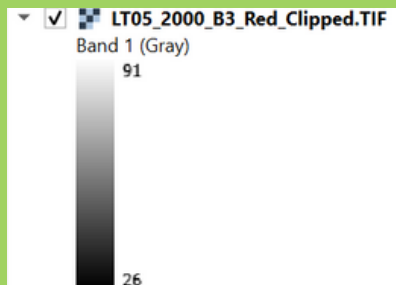
Downloaded Landsat images (2000, 2023)



2000 B3, B4

2023 B4, B5

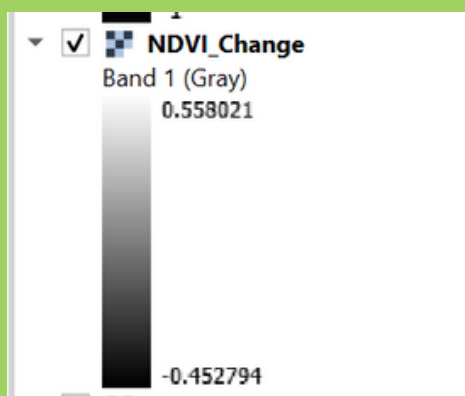
Computed NDVI using Raster Calculator



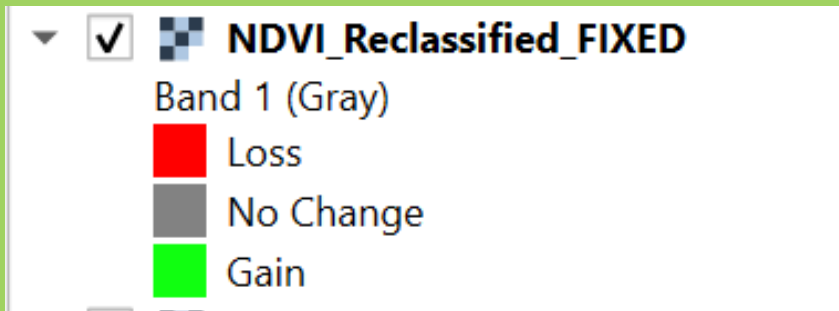
Clipped to AOI boundary



Created difference raster (2023 - 2000)



Reclassified into Gain, Loss, No Change



Converted raster to vector polygons



Calculated area statistics

fid	2
class	-1
count	7770
unique	7770
min	0.08998397556965938
max	437.4376106227487
range	437.347626647179
sum	5150.932981748894
mean	0.6629257376768204
median	0.179992428680812
stddev	6.098030550271234
minority	0.08998397556965938
majority	0.08998397556965938
q1	0.0900089515110245
q3	0.3600147549301153
iqr	0.2700058034190908

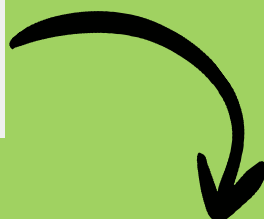
class = -1

fid	3
class	0
count	3450
unique	3450
min	0.08998312268198934
max	429920.1166014229
range	429920.02661830024
sum	431008.88929611153
mean	124.93011283945262
median	0.09004705426598084
stddev	7318.379939639626
minority	0.08998312268198934
majority	0.08998312268198934
q1	0.09002195515127387
q3	0.2700781580453273
iqr	0.18005620289405344

class = 0

fid	1
class	1
count	70053
unique	70053
min	0.08998357682966161
max	1396.0803369930431
range	1395.9903534162136
sum	46680.851572673884
mean	0.666364774851525
median	0.18000646068113857
stddev	8.306040904191489
minority	0.08998357682966161
majority	0.08998357682966161
q1	0.09002557214079425
q3	0.3601028635062277
iqr	0.27007729136543346

class = 1



Prepared final map layout



Methodology

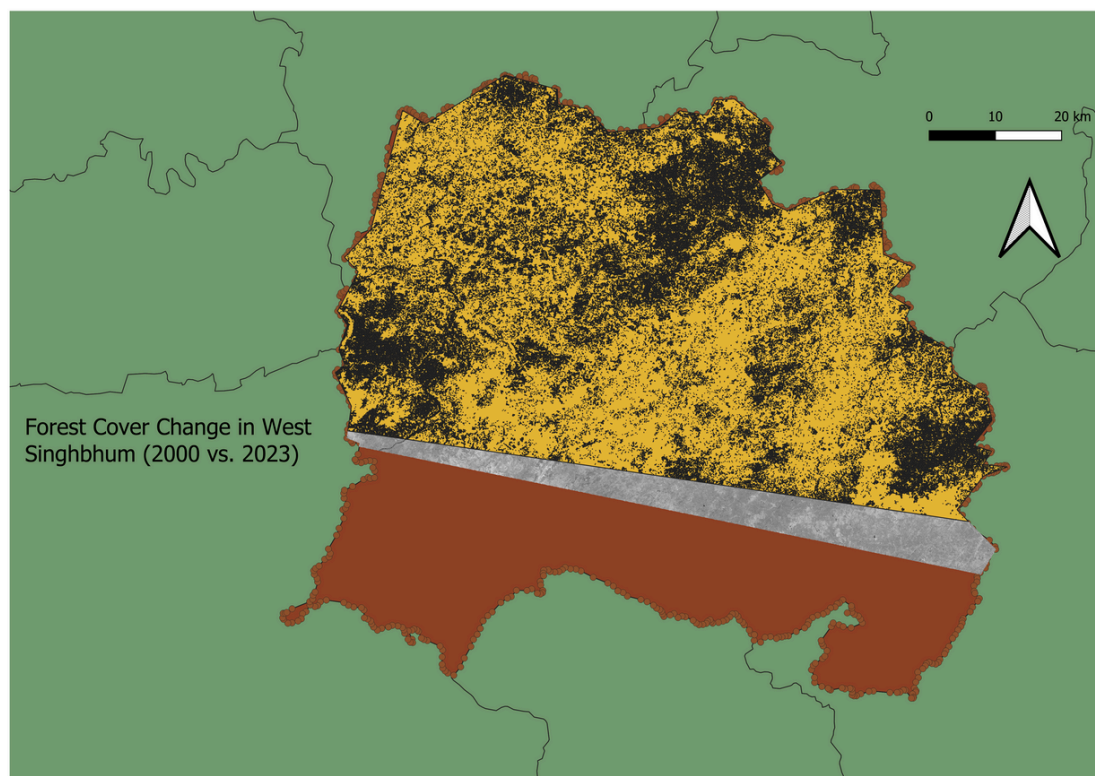
Landsat satellite imagery was acquired for the two time periods. The Red and Near-Infrared bands were used to calculate NDVI values for each year, generating "greenness" maps of the area. The core of the analysis involved creating a change detection raster by subtracting the 2000 NDVI values from the 2023 values.

This resulting difference map was then reclassified into three distinct categories: Vegetation Loss, No Significant Change, and Vegetation Gain. Finally, this classified raster was converted into vector polygons to calculate the precise area statistics for each change category.

in the next slide we can see the results that we obtained after applying the methodology above.

Results

NDVI_Change_Polygons_NEW
NDVI_Reclassified_FIXED
Band 1 (Gray)
Loss
No Change
Gain
NDVI_Reclassified
Band 1 (Gray)
1
-1
NDVI_Change
Band 1 (Gray)
0.558021
-0.452794
NDVI_2023
Band 1 (Gray)
0.455173
-0.213109
NDVI_2000
Band 1 (Gray)
0.495495
-0.349593
LC08_2023_B5_NIR_Clippped
Band 1 (Gray)
25,669
7,187
west singhbhum_vertices
west singhbhum_boundary
LT05_2000_B4_NIR_Clippped.TIF
Band 1 (Gray)
96
21
LT05_2000_B3_Red_Clippped.TIF
Band 1 (Gray)
91
26
vertices_simplified
west singhbhum_boundary.shp
WestSinghbhum_Boundary — gadm41_ind_2
gadm41_IND_2



	fid	class	count	unique	min	max	range	sum
1	1	1	70053	70053	0.08998357682...	1396.08033699...	1395.99035341...	46680.8515726...
2	2	-1	7770	7770	0.08998397556...	437.437610622...	437.347626647...	5150.93298174...
3	3	0	3450	3450	0.08998312268...	429920.116601...	429920.026618...	431008.889296...

mean	median	stddev	minority	majority	q1	q3	iqr
0.66636477485...	0.18000646068...	8.30604090419...	0.08998357682...	0.08998357682...	0.09002557214...	0.36010286350...	0.27007729136...
0.66292573767...	0.17999242868...	6.09803055027...	0.08998397556...	0.08998397556...	0.09000895151...	0.36001475493...	0.27000580341...
124.930112839...	0.09004705426...	7318.37993963...	0.08998312268...	0.08998312268...	0.09002195515...	0.27007815804...	0.18005620289...

Conclusion

NDVI differencing is a powerful and widely-used method to detect changes in vegetation over time. This project gives participants hands-on experience in analyzing forest dynamics using spectral indices and remote sensing techniques. However, accurate analysis depends on good quality images (cloud-free and seasonal match). In regions with mixed land cover, combining NDVI with other indices or visual inspection enhances interpretation. This analysis is valuable for forestry departments, environmental researchers, and climate studies.

Our analysis using NDVI differencing revealed clear shifts in forest cover between 2000 and 2023:

- Forest Recovery: Approximately 63 km² of new vegetation grew, likely due to conservation efforts or natural regrowth.
- Deforestation: Around 7 km² of forest was lost, possibly from logging, mining, or land-use changes.

These insights help us in tracking the success of reforestation programs in Jharkhand.

They help us in finding areas that are at risk and need protection.

There can be certain limitations to this such as

Clouds obscuring satellite views.

Landsat's 30m resolution can miss small-scale changes.

We can improve this by using

sentinel 2 to spot finer details

or using Google Earth Engine for better analysis of larger dataset over time.

References

- USGS Earth Explorer:
<https://earthexplorer.usgs.gov/>
- GADM: <https://gadm.org/>